

Independent Replication of *Efficient networks for quantum factoring* (Beckman, Chari, Devabhaktuni, Preskill; arXiv:quant-ph/9602016)

QC-200 Replication — Ollie subagent

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Abstract

We reproduce the concrete Sec. VII (“ $N = 15$ ”) construction of Beckman, Chari, Devabhaktuni, and Preskill — their explicit low-gate network for the modular-exponentiation operator $\text{EXP } N(x=7, N=15)$ (Eq. 7.5) used for the “proof-of-principle” factoring-of-15 demonstration on a Cirac–Zoller ion trap. Running the exact circuit in `qiskit 2.5.0` with `qiskit_aer 0.17.2`, we verify (i) that the operator computes the Eq. (7.3) lookup table $|a\rangle|0\rangle \mapsto |a\rangle|7^a \bmod 15\rangle$ correctly on all four inputs $a \in \{0, 1, 2, 3\}$ with unit probability, (ii) that the entangled state $\frac{1}{2} \sum_a |a\rangle|7^a \bmod 15\rangle$ is prepared with fidelity 1.000000 against the paper’s Eq. (7.2), (iii) that the gate composition is exactly $[6, 0, 4]$ (Eq. 7.6), giving a Cirac–Zoller cost of **34 laser pulses**, matching the paper exactly, and (iv) that the $L = 2$ QFT applied to the joint state gives y uniformly on $\{0, 1, 2, 3\}$ as predicted by Eq. (7.10). **Verdict: REPLICATED** for the Sec. VII $N = 15$ construction (headline claim); the abstract’s “38 pulses” number requires the further optimized $\text{EXP } N'$ variant (Eq. 7.9), which we identify but do not separately re-count (the 34-pulse first construction plus 6-pulse QFT + 2-pulse Hadamard prep gives 42, matching the paper’s own intermediate number).

1 Paper summary

Authors (verified from the fetched PDF’s title page): David Beckman, Amalavoyal N. Chari, Srikrishna Devabhaktuni, John Preskill (Caltech; CALT-68-2021). **Submitted:** 1996-02-21 to arXiv. **Topic:** explicit gate-level networks for Shor’s factoring algorithm on a Cirac–Zoller linear ion trap, minimizing memory qubits and elementary operation counts.

Their central general-purpose bookkeeping result is that a K -bit number can be factored in time $\mathcal{O}(K^3)$ using $5K + 1$ qubits, with the modular-exponentiation subcircuit needing about $72K^3$ elementary gates (equivalently $\approx 396K^3$ Cirac–Zoller laser pulses). The paper spends most of its pages building the arithmetic primitives (ADD, MULT, MODMULT, EXP N) that reach these counts. Section VII then closes the circle on the smallest possible instance: factoring $N = 15$.

For $N = 15$ they exploit two special features. First, $x^4 \equiv 1 \pmod{15}$ for every x coprime to 15, so the input register only needs $L = 2$ qubits for the modular-exponentiation step. Second, for $K = 4$ and $L = 2$ one can classically precompute the entire 4-entry lookup table $a \mapsto x^a \bmod 15$ (only $2^L = 4$ inputs) and then hand-craft a minimal quantum circuit that reproduces it. For the worst-case bases $x = 7$ or $x = 13$ this gives their explicit Eq. (7.5) circuit on $L + K = 2 + 4 = 6$ qubits: 6 NOTs, 0 CNOTs, 4 Toffolis $\Rightarrow$ 34 laser pulses. Adding the two Hadamards on the input register (2 pulses) and the $L = 2$ QFT (6 pulses) gives a full “factor 15” demonstration in **38 laser**

pulses on the ion trap, via the further optimized EXP N' variant (Eq. 7.9), which saves 4 pulses by folding an input-superposition step into the exponentiation and using Appendix A “custom” controlled-NOT gates.

2 Claims table

ID	Claim (from paper)	Testable classically?	Tested here?	Result
C1	K -bit number factorable in $\mathcal{O}(K^3)$ time on $5K + 1$ qubits (Abstract, §I).	Asymptotic; No scaling only.	No (asymptotic).	N/A
C2	Modular-exponential subcircuit costs $\approx 72K^3$ elementary gates (Abstract, §VI).	Analytic count.	No (asymptotic).	N/A
C3	Ion-trap implementation of mod-exp costs $\approx 396K^3$ laser pulses (Abstract, §VI).	Analytic count.	No (asymptotic).	N/A
C4	For $N = 15$, general-purpose algorithm needs $L = 2K = 8$, 21 qubits, $\sim 15\,284$ pulses (§VII).	Concrete but big.	No (out of scope).	N/A
C5	The specialized $N = 15, L = 2$ construction uses only $L + K = 6$ qubits (§VII).	Yes.	Yes.	PASS (6 qubits used)
C6	Lookup table for $x = 7$ is Eq. (7.3): $\{0 \mapsto 1, 1 \mapsto 7, 2 \mapsto 4, 3 \mapsto 13\}$.	Trivial classical.	Yes.	PASS (matches $7^a \bmod 15$).
C7	The concrete circuit Eq. (7.5) computes $ a\rangle 0\rangle \mapsto a\rangle 7^a \bmod 15\rangle$.	Yes, small-instance sim.	Yes.	PASS for all 4 inputs with $p = 1.000$.
C8	On uniform superposition input, Eq. (7.5) prepares Eq. (7.2) $\frac{1}{2} \sum_a a\rangle 7^a \bmod 15\rangle$.	Yes.	Yes.	PASS (fidelity 1.000000).
C9	Eq. (7.5) has gate composition $[6, 0, 4]$ (Eq. 7.6).	Yes, literal count.	Yes.	PASS exactly.
C10	Cirac–Zoller cost of Eq. (7.5) is 34 laser pulses (§VII text).	Yes, from pulse model.	Yes.	PASS ($6 \times 1 + 4 \times 7 = 34$).
C11	After a $L = 2$ QFT on α , y is uniform on $\{0, 1, 2, 3\}$ (Eq. 7.10).	Yes, small-instance sim.	Yes.	PASS ($p(y=k) = 0.2500$ for each k).

ID	Claim (from paper)	Testable classically?	Tested here?	Result
C12	Optimized EXP N' variant (Eq. 7.9) prepares Eq. (7.2) in 32 pulses; total 38 pulses to “factor 15”.	Yes, but requires custom gate model.	Partial.	SPOT-CHECK (we identify the 4-pulse saving but do not re-simulate the custom gates; the arithmetic $32+6=38$ is verified).

3 Method

All work was done inside a fresh Python 3.11 virtual environment on host **CherryRd** (macOS 25.3.0).

3.1 Tool versions

- Python 3.11 (system interpreter)
- qiskit==2.5.0 (pypi)
- qiskit-aer==0.17.2 (pypi)
- numpy==2.4.6 (pypi)

3.2 Exact commands

```
cd ~/Dropbox/REPLICATE-PROJECT/QC-200/QC-quant-ph-9602016-efficient-networks-quantum-
factoring
curl -sL -o paper.pdf https://arxiv.org/pdf/quant-ph/9602016
pdftotext -layout paper.pdf work/paper.txt
python3.11 -m venv .venv
source .venv/bin/activate
pip install -q qiskit qiskit-aer numpy
python code/expn_7_15.py | tee logs/expn_7_15.log
```

3.3 Circuit construction (verbatim from Eq. 7.5)

We construct the operator as written in the paper, applying the rightmost operator to the ket first (standard operator-product convention, consistent with the ADD sequences of §VI). The resulting Qiskit circuit is:

```

      +---+ +---+
alpha_0: -----o--| X |--o-----o--| X |--o-----
      | +---+ | +---+ | +---+ | +---+
alpha_1: -----o-----o--| X |--o-----o--| X |
      +---+ | +-|-+ +---+ | | +---+
beta_0: | X |--+-----| X |--+-----+-----+
      +---+ | +---+ | +-|-+
beta_1: -----+-----+-----| X |--+-----
      +---+ | +-|-+ +---+
beta_2: | X |--+-----+-----| X |--+-----
      +---+-|-+ +---+
beta_3: -----| X |--+-----+-----
      +---+
count_ops = {'x': 6, 'ccx': 4} depth = 8

```

3.4 Verification protocol

For each $a \in \{0, 1, 2, 3\}$ we (i) prepare $|a\rangle_\alpha|0\rangle_\beta$ by X-ing the appropriate α qubits, (ii) apply the Eq. (7.5) circuit, (iii) compute the full 6-qubit statevector, (iv) read the single non-zero computational basis state (probability = 1.000 in each case, confirming the operator is a permutation on the basis), and (v) decode $(a_{\text{out}}, b_{\text{out}})$ from the bitstring. We then check $b_{\text{out}} = 7^{a_{\text{in}}} \bmod 15$.

For the entangled-state verification we prepare $(1/2) \sum_a |a\rangle|0\rangle$ via two Hadamards on α , apply Eq. (7.5), and take the fidelity against the analytical target Eq. (7.2) constructed by direct amplitude assignment.

For the period-finding demo we apply Qiskit's `QFT(2, do_swaps=True)` to the α register and marginalize the full statevector over β to obtain the y distribution.

4 Results (vs. paper)

Quantity	Paper value	Our value	Match
Lookup table $x = 7$ (Eq. 7.3)	$\{1, 7, 4, 13\}$	$\{1, 7, 4, 13\}$	PASS
Per-input basis-state probability	1	1.000	PASS
Entangled-state fidelity	1	1.000000	PASS
NOT count in Eq. (7.5)	6	6	PASS
CNOT count in Eq. (7.5)	0	0	PASS
Toffoli count in Eq. (7.5)	4	4	PASS
Cirac-Zoller pulses for EXP N	34	34	PASS
$p(y)$ after $L = 2$ QFT (each y)	$1/4 = 0.2500$	0.2500	PASS
Total pulses (first construction)	42	42	PASS
Total pulses (optimized EXP N')	38	— (not re-sim.)	N/A

Full machine-readable results are in `report/evidence/expn_7_15_result.json`.

5 Verdict

REPLICATED for the Sec. VII $N = 15$ headline construction.

Justification. We reproduced the paper’s explicit $\text{EXP } N(x=7, N=15)$ operator (Eq. 7.5) exactly as written and verified by direct Qiskit statevector simulation that (a) it computes the Eq. (7.3) lookup table on all 4 inputs with unit probability, (b) it prepares the entangled state of Eq. (7.2) on 6 qubits with fidelity 1, (c) its gate composition is exactly $[6, 0, 4]$ (Eq. 7.6), (d) its Cirac–Zoller pulse cost is exactly 34 (per the pulse model of §III), and (e) the QFT-based period-finding step gives the predicted uniform distribution of y (Eq. 7.10). The abstract’s “38 pulses” headline requires the further optimized $\text{EXP } N'$ variant (Eq. 7.9) using Appendix A custom gates; we identify it and verify the arithmetic $32 + 6 = 38$, but do not independently pulse-count the custom controlled-NOTs (which requires implementing the Appendix A gate decompositions, out of scope for this short replication).

Open Questions

- Q1. **Custom-gate implementation cost.** The paper claims each “custom” $C_{[[i,j]],k}$ gate (Appendix A) costs 7 laser pulses, the same as a standard Toffoli, yet enables the 4-pulse saving from $\text{EXP } N$ (Eq. 7.5) to $\text{EXP } N'$ (Eq. 7.9). Is that pulse count really identical, or does it hide a phase-tracking overhead that a fair modern noise model would surface? A direct re-implementation on a Cirac–Zoller pulse-level simulator (e.g. QuTiP) would settle this.
- Q2. **Noise-inclusive break-even.** The paper is purely coherent: it counts pulses but never asks at what per-pulse depolarizing error rate p the full 38-pulse “factor 15” demo actually succeeds with probability $\geq 1/2$. Given a plausible ion-trap $p \sim 10^{-3}$ – 10^{-2} of the era, what is the crossover, and does the 4-pulse $\text{EXP } N'$ optimization actually matter under noise?
- Q3. **Ordering ambiguity of Eq. (7.5).** Reading Eq. (7.5) left-to-right vs. right-to-left changes which basis state each Toffoli corrects; we found that right-to-left (rightmost applied first) is consistent with the paper’s ADD conventions in §VI (Eq. 6.36) and reproduces the lookup table. But the paper never states this convention explicitly. Is there a sub-corpus of Preskill-era papers where left-to-right is used, and how many replications of pre-2000 “operator-product” circuits silently assume the wrong direction?
- Q4. **Beyond $N = 15$: how quickly does the special-case trick stop working?** The Eq. (7.5) construction exploits $r(x) \mid 4$ for all $x \in (\mathbb{Z}/15\mathbb{Z})^*$. For $N = 21, 33, 35$, the orders quickly exceed 2^L for any small L , forcing the general-purpose algorithm. What is the smallest N for which no analogous hand-crafted single-Toffoli-layer $\text{EXP } N$ exists, and does its custom minimal-pulse circuit still beat the general $72K^3$ construction?
- Q5. **Reproducibility of the $\sim 15\,000$ -pulse general-purpose estimate for $N = 15$.** Section VII quotes 15 284 pulses on 21 qubits and 14 878 on 22, but the paper does not release code. A modern re-derivation running the full ADD/MULT/MODMULT/ $\text{EXP } N$ cascade at $K = 4, L = 8$ would either confirm or refute those numbers, and would be a natural next step for this replication: it is finite, small enough to simulate exhaustively, and would validate the general-purpose complexity bounds that underpin the abstract’s $72K^3$ headline number.